SOLUTION SHEET 2 - Conditional Programming

1.

```
a = int(input("Enter your number: "))
if a % 7 ==0 :
   print("It is divisible.")
else:
   print("It is not divisible.")
```

2.

```
a = int(input("Enter your number: "))
if a % 10 ==4 :
   print("It is true.")
else:
   print("It is false.")
```

3.

```
a = int(input("Enter your number: "))
if a % 10 ==4 and a % 3 ==0:
   print("It is true.")
else:
   print("It is false.")
```

4.

```
a = int(input("Enter your number: "))
if a % 10 ==7 or a % 5 ==0:
   print("It is true.")
else:
   print("It is false.")
```

5.

```
a = int(input("Enter your number: "))
if a % 5 == 0 and a % 11 ==0:
    print("It is true.")
else:
    print("It is false.")
```

6.

```
a = int(input("Enter num1: "))
b = int(input("Enter num2: "))
c = int(input("Enter num3: "))
if a > b and a > c:
    print("a is maximum")
elif b > c:
    print("b is maximum")
else:
    print("c is maximum")
```

7.

```
a = int(input("Enter angle1: "))
b = int(input("Enter angle2: "))
c = int(input("Enter angle3: "))
if a + b == c or  a + c == b or b +c == a:
    print("It is an obtuse triangle")
elif a == 90 or b == 90 or c == 90:
    print("It is a right agnle triangle")
elif a == b ==c == 60:
    print("It is an equilateral triangle")
else:
    print("It is an acute triangle")
```

8.

```python
age = int(input("Enter your age: "))
if age>=18:
    print("You're eligible to vote.")
else:
    print("You're not eligible to vote.")
```

9.

```python
year = int(input("Enter the year: "))
if year % 400 == 0 or year % 100 != 0 and year % 4 == 0:
    print("It is a leap Year")
else:
    print("It is not leap year")
```

10.

```python
day = int(input("Enter your number (1-7): "))
if day == 1:
    print("Monday")
elif day == 2:
    print("Tuesday")
elif day == 3:
    print("Wednesday")
elif day == 4:
    print("Thursday")
elif day == 5:
    print("Friday")
elif day == 6:
    print("Saturday")
elif day == 7:
    print("Sunday")
else:
    print("Wrong value")
```

11.

```
A,B,C,D,E = int(input("Enter num1: ")),int(input("Enter num2: ")), int(input("Enter num3: ")),
int(input("Enter num4: ")), int(input("Enter num5: ")),

print("Your average is: ")

print((A+B+C+D+E)/5)
```

12.

```
A,B,C = int(input("Enter angle1: ")), int(input("Enter angle2: ")), int(input("Enter angle3: "))

if A + B + C == 180:

   print("Valid triangle")

else:

   print("Triangle invalid")
```

13.

```
a = int(input("Enter num1: "))

b = int(input("Enter num2: "))

if a > b:

   print("a is maximum")

else:

   print("b is maximum")
```

14.

```
a = int(input("Enter num1: "))

b = int(input("Enter num2: "))

c = int(input("Enter num3: "))

if a < b and a < c:

   print("a is minimum")

elif b < c:

   print("b is minimum")

else:
```

```python
    print("c is minimum")
```

15.

```python
percent = int(input("Enter your percentage: "))
if percent < 25 and percent > 0:
    print("D grade")
elif percent > 25 and percent < 45:
    print("C grade")
elif percent > 45 and percent < 65:
    print("B grade")
elif percent > 65 and percent < 85:
    print("A grade")
elif percent > 85 and percent < 100:
    print("A+ grade")
else:
    print("Invalid percentage")
```

SOLUTION SHEET 3 - Control Statement (Loops)

1.

```
n = int(input("Enter the num: "))
i = 1
while i<=n:
    print(i, end=" ")
    i+=1
```

2.

```
n = int(input("Enter the num: "))
i = 1
while i<=n:
    print(n, end=" ")
    n-=1
```

3.

```
n = int(input("Enter the num: "))
i = 1
while i<=n:
    if i %2 ==0:
        print(i)
    i+=1
```

4.

```
n = int(input("Enter the num: "))
i = 1
while i<=n:
    if i %2 !=0:
        print(i, end=" ")
    i+=1
```

5.

```
n = int(input("Enter the num: "))
i = 1
numSum = 0
while i<=n:
   numSum=numSum+i
   i+=1
print(numSum)
```

6.

```
A = int(input("Enter the num: "))
i = 1
numSum = 0
while i<=A:
   if i % 2 == 0:
      numSum=numSum+i
   i+=1
print(numSum)
```

7.

```
A = int(input("Enter the num: "))
i = 1
numSum = 0
while i<=A:
   if i %2 != 0:
      numSum=numSum+i
   i+=1
print(numSum)
```

8.

```
N = int(input("Enter the num: "))
i = 1
times = 1
while i == 1:
    if N // 10 !=0:
        N = N // 10
        times+=1
    else:
        i = 0
print(times)
```

9.

```
N = int(input("Enter the num: "))
numSum = 0
while N > 0:
    digit = N % 10
    numSum = numSum + digit
    N = N // 10
print(numSum)
```

10.

```
A = int(input("Enter the num: "))
original_A = A
reversed_A = 0
while A > 0:
    digit = A % 10
    reversed_A = reversed_A * 10 + digit
    A = A // 10
if original_A == reversed_A:
    print("Yes")
```

```
else:
   print("No")
```

11.

```
A = int(input("Enter the num: "))
for i in range(10):
   table = (i+1)*A
   print(A,"*", i + 1, "=", table)
   i+=1
```

12.

```
A = int(input("Enter the num1: "))
B = int(input("Enter the num2: "))
power = 1
for i in range(B):
   power = power * A
print(power)
```

SOLUTION SHEET 4 - PATTERN PRINTING

1.

```
for i in range(4):
    for j in range(5):
        print("*", end="")
    print()
```

2.

```
for i in range(5):
    for j in range(i+1):
        print("*", end="")
    print()
```

3.

```
for i in range(5,0, -1):
    for j in range(i):
        print("*", end="")
    print()
```

4.

```
for i in range(5):
    for j in range(i +1):
        if j % 2 == 0:
            print(j+1, end="")
        else:
            print("*", end="")
    print()
```

5.

```
N = 5
```

```
for i in range(4):
    for j in range(N):
        if j == 0 or j == N-1:
            print("*", end="")
        else:
            print("_", end="")
    print()
```

6.

```
for i in range(6,1, -1):
    N  = i+1
    for j in range(N):
        if j ==0 or j == N-1:
            print("*", end="")
        else:
            print("_", end="")
    print()
```

7.

```
for i in range(5,0,-1):
    for j in range(5):
        if i <= j+1:
            print("*", end="")
        else:
            print("_", end="")
    print()
```

8.

```
for i in range(0,5):
    for j in range(0,5):
        if i <= j:
```

```
            print("*", end="")
        else:
            print("_", end="")
    print()
```

9.

```
rows = int(input("rows: "))
columns = (rows * 2)
value= rows - 1
increment = rows

for i in range(rows):
    for j in range(columns):
        if i + j > value and i + j < increment:
            print(" ", end="")
        else:
            print("*", end="")
    increment+=2
    print('')
```

10.

```
rows = 5
columns = 10
increment = 1
decrement = columns
for i in range(rows):
    for j in range(columns):
        if i + j < increment or i + j + 1 > decrement:
            print("*", end="")
        else:
            print(" ", end="")
```

```
    increment += 2
    print('\n')
```

11.

```
rows = 5
columns = 9
increment = 5
decrement = 1
for i in range(0, rows):
    for j in range(0, columns+1):
        if i + j > 4  and i + j < increment:
            print(" ", end="")
        else:
            print("*", end="")
    increment+=2
    print()
r = 5
c = 9
inc = 1
dec = c
for i in range(0, r):
    for j in range(0, c+1):
        if i + j < inc or i + j + 1 > dec:
            print("*", end="")
        else:
            print(" ", end="")
    inc+=2
    print()
```

12.

```
for i in range(4):
```

```
    for j in range(i+1):
        print(j+1, end="")
    print()
```

13.

```
for i in range(4,0,-1):
    for j in range(i):
        print(j+1, end="")
    print()
```

14.

```
N = 1
for i in range(1,5):
    for j in range(0, i):
        print(N, end="")
        N+=1
    print()
```

15.

```
for i in range(5):
    for j in range(i+1):
        print("*", end="")
    print()

for m in range(6,0, -1):
    for n in range(m):
        print("*", end="")
    print()
```

SOLUTION SHEET 5 - LIST

1.

```
arr = list(map(int, input().split()))
arrSum = 0
for i in range(len(arr)):
   arrSum += arr[i]
print(arrSum)
```

2.

```
A = list(map(int, input().split()))
B = 3
arr = []
for i in range(len(A)):
   arr.append((A[i] + B))
print(arr)
```

3.

```
arr = list(map(int, input().split()))
maxVal = arr[0]
minVal = arr[0]
for i in range(1, len(arr)):
   if arr[i] > maxVal:
      maxVal = arr[i]
   if arr[i] < minVal:
      minVal = arr[i]

print(maxVal, minVal)
```

4.

```
arr = list(map(int, input().split()))
target = 3
```

```
for i in range(len(arr)):
    if arr[i] == target:
        print("Value found at index: ", end="")
        print(i)
    else:
        print("Not found")
```

5.

```
arr = list(map(int, input().split()))
for i in range(len(arr)):
    if arr[i]<0:
        print(arr[i])
```

6.

```
A = list(map(int, input().split()))
even, odd = 0, 0
for num in A:
    if num % 2 == 0:
        even += 1
    else:
        odd += 1
absolute = abs(even - odd)
print(absolute)
```

7.

```
arr = list(map(int, input().split()))
even = []
odd = []
for i in range(len(arr)):
    if arr[i] % 2 == 0:
        even.append(arr[i])
```

```
    else:
        odd.append(arr[i])
print(even)
print(odd)
```

8.

```
arr = list(map(int, input().split()))
newArr= []
for i in range(len(arr)):
    newArr.append(arr[i]**2)
print(newArr)
```

9.

```
arr = list(map(int, input().split()))
newArr= []
for i in range(len(arr)):
    newArr.append(arr[i]**3)
print(newArr)
```

10.

```
arr = [1,2,3,4,5]
N = len(arr)
for i in range(N//2):
    temp = arr[i]
    arr[i] = arr[N-i-1]
    arr[N-i-1] = temp
print(arr)
```

11.

```
arr = [1,2,3,4,5]
N = len(arr)
```

```
for i in range(N//2):
    temp = arr[i]
    arr[i] = arr[N-i-1]
    arr[N-i-1] = temp
print(arr)
```

12.

[2, 4, 6, 8, 10, 12, 14, 16, 18, 20]

[2, 4, 6, 8, 10, 12, 14, 16, 18, 20]

[6, 8, 10]

[6, 8, 10, 12, 14, 16, 18, 20]

[6, 8, 10, 12, 14, 16, 18, 20]

[2, 4]

[2, 6, 10, 14, 18]

[4, 8, 12, 16, 20]

[6, 10, 14, 18]

13.

11

[19, 17, 15]

[19, 21]

[19, 21]

[1, 3, 5, 7, 9, 11, 13, 15, 17]

[21, 17, 13, 9, 5, 1]

[21, 19, 17, 15, 13, 11, 9, 7, 5, 3, 1]

14.

['Nikhil', 5, 4, 3, 2, 1.4]

15.

['Hello', 'everyone', 'how', 'are', 'you']

['Hello', 'everyone', 'how are you']

['Nikhil', 'Chauhan', 'Delhi']

['23456']

['2', '3', '4', '5']

16.

[1, 2, 3, 5, 8, 9, 3, 4, 5, 6, 7, 10]

[1, 2, 3, 5, 8, 9, 1, 2, 3, 5, 8, 9, 1, 2, 3, 5, 8, 9]

17 (Miscellaneous).

```
arr = [2,2,3,3,4,4,5,5,6,6,7,7,9,8,8]
N = len(arr)
value = 0
for i in range(N):
    target = arr[i]
    count = 0
    for j in range(N):
        if target == arr[j]:
            count+=1
    if count == 1:
        value = arr[i]
print(value)
```

SOLUTION SHEET 6 - STRINGS

1.

```
t = int(input("Enter num of strings: "))
vow = 'aeiou'
l = []
for i in range(t):
   l.append(input().lower())
for s in l:
   v = c = 0
   for char in s:
      if char in vow:
         v += 1
      else:
         c += 1
   print(v, c)
```

2.

```
a = "hello"
myList = list(a)
print(len(myList))
```

3.

```
t = int(input("Enter num of strings: "))
l = []
for i in range(t):
   l.append(input().lower())
for i in l:
   if i == i[::-1]:
      print(1)
   else:
      print(0)
```

4.

```
a = "**h*e*l*lo"
print(a.strip("*"))
```

5.

```
a = "**h*e*l*lo"
print(a.lstrip("*"))
```

6.

```
a = "**h*e*l*lo"
print(a.rstrip("*"))
```

7.

```
a = "String"
print(a[::-1])
```

8.

```
a = "Suyash Chaudhary"
new = a.split()
new.reverse()
st = " ".join(new)
print(st)
```

9.

```
a = "Everyone loves data science"
new = a.split()
myL = []
for i in range(len(new)):
  myL.append(new[i][::-1])
print(' '.join(myL))
```

10.

```
a = "PythoN"
print(a.lower())
```

11.

```
a = "pYthON"
print(a.upper())
```

12.

```
A = "Python45"
if A.isalpha():
    print(1)
else:
    print(0)
```

```
A = "Python45"
if A.isalnum():
    print(1)
else:
    print(0)
```

13.

```
A = "aabbcc"
B = 98
new = chr(B)
print(A.find(new))
```

14.

```
A = "aababbabaa"
B = "ba"
```

```
if B in A:
    print(A.find(B))
```

15.

```
A = "bobob"
length_A = len(A)
count_bob = 0
for i in range(length_A - 2):
    if A[i:i+3] == "bob":
        count_bob += 1
print(count_bob)
```

16.

```
A= "aeiOUz"
A+=A
vow = 'aeiou'
result =''
for i in range(len(A)):
    if A[i] in vow:
        result += '#'
    if A[i].islower() and A[i] not in vow:
        result+=A[i]
print(result)
```

SOLUTION SHEET 7 - DICTIONARY

1.

```
A = [2, 6, 3, 8, 2, 8, 2, 3, 8, 8]

B = 2

freq_dict = {}

for num in A:

    if num in freq_dict:

        freq_dict[num] += 1

    else:

        freq_dict[num] = 1

print(freq_dict.get(B, 0))
```

2.

```
A = [1, 2, 3, 1, 2, 5]

non_repeating = None

for num in A:

    if A.count(num) == 1:

        non_repeating = num

        break

print(non_repeating)
```

3.

```
A = "abcde"

char_freq = {}

for char in A:

    char_freq[char] = char_freq.get(char, 0) + 1

odd_count = sum(1 for count in char_freq.values() if count % 2 != 0)

print(int(odd_count <= 1))
```

4.

```
A = [10, 5, 3, 4, 3, 5, 6]
```

```
seen = {}
for num in A:
   if num in seen:
      print(num)
      break
   seen[num] = 1
else:
   print(-1)
```

5.

```
dict1 = {'Ten': 10, 'Twenty': 20, 'Thirty': 30}
dict2 = {'Thirty': 30, 'Fourty': 40, 'Fifty': 50}
for key, value in dict2.items():
   dict1[key] = value
print(dict1)
```

6.

```
dict1 = {'a': 100, 'b': 400, 'c': 300}
value = 400
found = False
for val in dict1.values():
   if val == value:
      found = True
      break
print(f"{value} present in a dict" if found else f"{value} not present in a dict")
```

7.

```
person = {"name": "abc", "age": 25}
for key in person.keys():
   print(key)
```

8.

```
person = {"name": "abc", "age": 25}
for value in person.values():
    print(value)
```

9.

```
person = {"name": "abc", "age": 25}
for key, value in person.items():
    print(key, value)
```

10.

```
person = {"name": "abc", "age": 25}
person.clear()
print(person)
```

11.

```
person = {"name": "abc", "age": 25}
keys_list = list(person.keys())
print(keys_list)
```

12.

```
person = {"name": "abc", "age": 25}
values_list = list(person.values())
print(values_list)
```

13.

```
N = 5
squares_dict = {x: x*x for x in range(1, N+1)}
print(squares_dict)
```

14.

```
dict1 = {1: 'a', 2: 'b', 3: 'c'}
list_of_tuples = [(key, value) for key, value in dict1.items()]
print(list_of_tuples)
```

15.

```
dict1 = {2: 'Apple', 1: 'Mango', 3: 'Orange', 4: 'Banana'}
sorted_keys = sorted(dict1.keys())
for key in sorted_keys:
    print(key, dict1[key])
```

16.

```
A = [1, 2, 2, 1]
B = [2, 3, 1, 2]
common_elements = [x for x in A if x in B]
print(common_elements)
```

17.

```
squares = [x*x for x in range(1, 6)]
print(squares)
```

18.

```
numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
even_numbers = [x for x in numbers if x % 2 == 0]
print(even_numbers)
```

19.

```
text = "hello"
uppercase_chars = [char.upper() for char in text]
print(uppercase_chars)
```

20.

```
a = 4

print(type(a))  # <class 'int'>

a = 4,

print(type(a))  # <class 'tuple'>

a = (4)

print(type(a))  # <class 'int'>

a = ()

print(type(a))  # <class 'tuple'>

a = (4,)

print(type(a))  # <class 'tuple'>

a = (4, 5)

print(type(a))  # <class 'tuple'>
```